

## Week 5 Assignment

Total marks=10

Q1. Calculate the transport number of  $\text{Li}^+$  and  $\text{Br}^-$  ions when a current flows through an infinitely dilute aqueous solution of  $\text{LiBr}$  at room temperature ( $25^\circ\text{C}$ ). Given the ionic mobilities of  $\text{Li}^+$  and  $\text{Br}^-$  at infinite solution are  $4.01 \times 10^{-8}$  and  $6.04 \times 10^{-8} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ , respectively

- a) 0.5723, 0.4277
- b) 0.60, 0.40
- c) 0.40, 0.60
- d) 0.669, 0.331

(2)

Q2. The cell constant of a conductivity cell

- a) Changes with change of electrolyte
- b) Changes with change of concentration of electrolyte
- c) Changes with temperature of electrolyte
- d) Remains constant for a cell

(1)

Q3. Conductivity of  $0.056 \text{ M Ag}_2\text{SO}_4$  is  $0.0028 \text{ mho cm}^{-1}$ . What happens to this conductivity with dilution

- a) Increases
- b) Decreases
- c) Remains unchanged
- d) First increases then decreases

(2)

Q4. The units of cell constant, specific conductance, conductance are

- a)  $\text{cm}$ ,  $\text{mho}^{-1} \text{cm}^{-1}$ , and  $\text{mho}$  respectively
- b)  $\text{cm}^{-1}$ ,  $\text{mho cm}^{-1}$ , and  $\text{mho}$  respectively
- c)  $\text{cm}^2$ ,  $\text{mho cm}^{-1}$ , and  $\text{mho}$  respectively
- $\text{mho cm}^{-1}$ ,  $\text{cm}^{-1}$ , and  $\text{mho cm}$  respectively

(2)

Q5. A solution containing 3.94 percent of copper sulphate was electrolyzed using a platinum cathode and a copper anode. After electrolysis the weight of the cathode was found to be 65.79 g and it contains 0.809g of copper and the increase in the weight of the cathode was 0.806g. Calculate the transport number of  $\text{Cu}^{2+}$

(Equivalent mass of Copper=31.8)

- a) 0.274
- b) 0.726
- c) 0.477
- d) 0.522

~~Difference = 0.809 - 0.806 = 0.003g~~  
~~this is migration of ions~~  
 $t_{\text{Cu}^{2+}} = \frac{\text{Loss in conc}}{\text{Total copper liberated}}$   
 $\therefore \text{Cu deposition} = \frac{0.806}{31.8}, 0.025 \text{ g} \quad (2)$

Q6. As temperature decreases electrolytic conduction

- a) increases
- b) decreases
- c) remains unaffected
- d) first increases then decreases

$\text{diff} = 0.809 - 0.806 = 0.003 \text{ g}$   
 $t_{\text{Cu}^{2+}} = \frac{\text{Loss in conc}}{\text{Total copper liberated}}$   
 $t_{\text{Cu}^{2+}} = \frac{0.809 - 0.806}{0.809}$   
 But copper used as anode (1)  
 $\therefore t_{\text{Cu}^{2+}} = 1 - \frac{0.806}{0.809}$   
 $t_{\text{Cu}^{2+}} = 0.726$